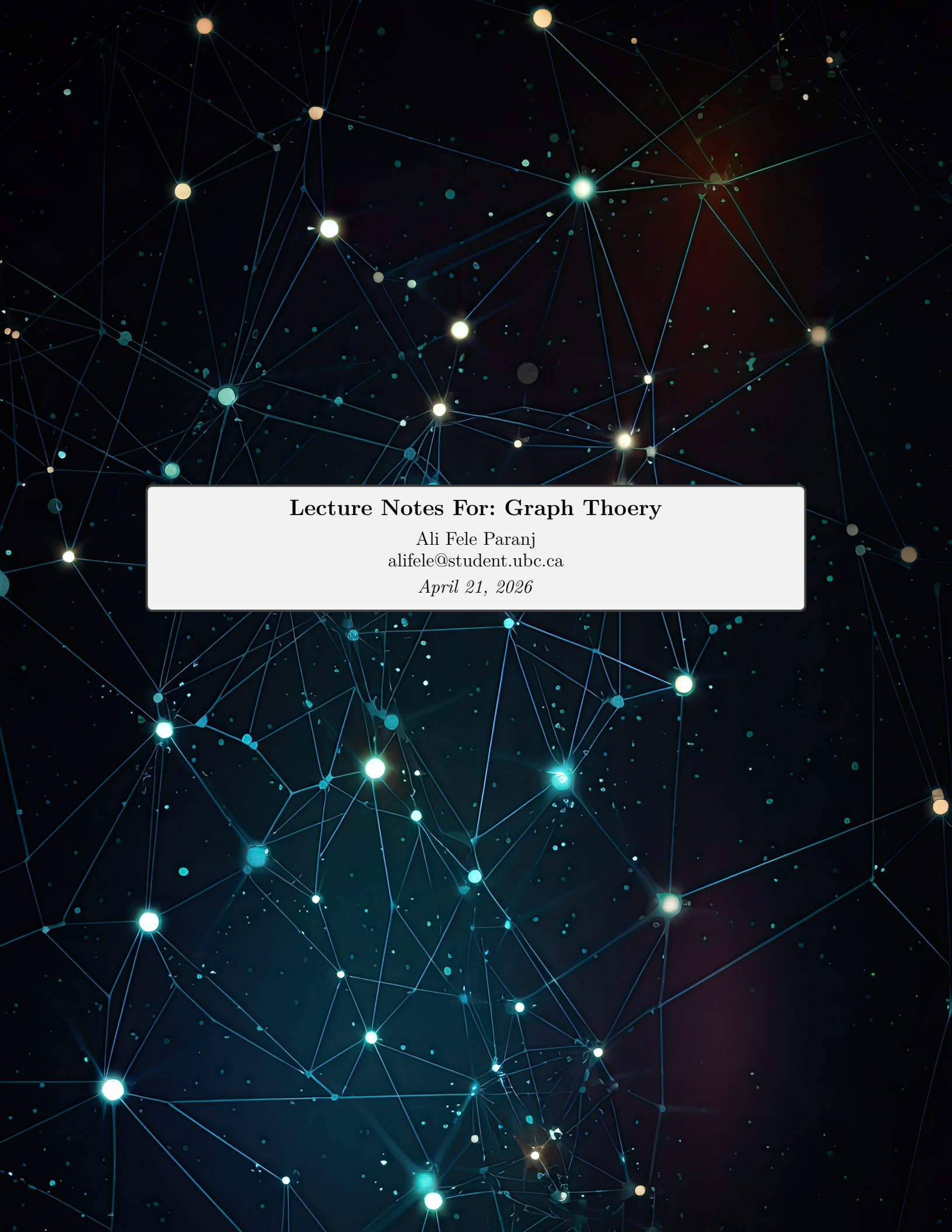

Lecture Notes For: Graph Thoery

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1. Basics and Definitions

Here in this chapter we will cover the sporadic definitions, theorems, and proofs for the graph theory to get ready for the upcoming chapters. So expect a high diversity and less coherency for the material covered in this chapter.

1.1 Basic Definitions and Notions of Graph Theory

Definition 1.1 The complement of graph $G = (V, E)$ is defined to be

$$\bar{G} = (V, E^c) = (V, \binom{V}{2} - E).$$

The following definition happens to be one of the central definitions in the realm of graph theory.

Definition 1.2 Two graphs G, H are isomorphic if there exists a bijection $\varphi : V(G) \to V(H)$ such that it preserves the edges, i.e.

$$xy \in E(G) \implies \varphi(x)\varphi(y) \in E(H).$$

The intuitive description of isomorphism between two graphs is that we can change one to the other one by just moving around the vertices and not connecting/disconnecting any edges. Combining the notion of the complement of a graph with the notion of isomorphic graphs we can have the following theorem.

Theorem 1.1 Two graphs are isomorphic if and only of their complement are isomorphic.

Proof. This proof will have two direction. For the forward direction, Let G and H be two isomorphic graphs (i.e. there exists a structure preserving bijection φ between the vertex set of two graphs). Let xy be a non-edge in G. I.e. $xy \in E(\bar{G})$ or equivalently $xy \notin E(G)$. Then it follows that $\varphi(x)\varphi(y) \notin E(H)$ which is equivalent to $\varphi(x)\varphi(y) \in E(\bar{H})$. Thus we can conclude that the same φ is a structure preserving map from $\bar{G}$ to $\bar{H}$ as well, thus $\bar{G}$ and $\bar{H}$ are isomorphism as well. The proof for the converse is very similar to the structure of the proof presented above. $\square$

1.2 What is a Graph

■ **Problem 1.1** As in the text book.

Solution $k_{1,1}$ is the only complete bipartite graph that is also complete.

■ **Problem 1.2** As in the textbook.

Solution The adjacency matrices are:

$$A_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad A_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

And the corresponding incidence matrices are:

$$M_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad M_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$$

An adjacency matrix for a path with six vertices is:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

And an adjacency matrix for a cycle with six vertices is:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

■ **Problem 1.3** As in the textbook.

Solution An adjacency matrix for $K_{n,m}$ can be written as

$$A = \begin{pmatrix} 0_{n\times n} & I_{n\times m} \\ I_{m\times n} & 0_{m\times m}, \end{pmatrix}$$

for which we have ordered the vertices as $(v_1, v_2, \cdots, v_n, v_{n+1}, \cdots, v_{n+m})$, where $v_1, \cdots, v_m$ all belong to one partide, and $v_{n+1}, \cdots, v_{n+m}$ all belong to the other partide.

■ **Problem 1.4** From the definition of isomorphism, prove that $G \cong H$ if and only if $\overline{G} \cong \overline{H}$.

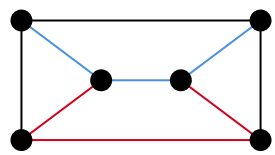
Solution For the first direction, assume $G \cong H$. We want to show $\overline{G} \cong \overline{H}$. Since $G \cong H$, $\exists f : V(G) \to V(H)$ such that $ab \in E(G)$ iff $f(a)f(b) \in E(H)$. Let $xy \in E(\overline{G})$. Then $xy \notin E(G)$. Using the fact that f is isomorphism, $f(x)f(y) \notin E(H)$. So $f(x)f(y) \in E(\overline{H})$. Furthermore, let $uv \notin E(\overline{G})$. So $uv \in E(G) \Leftrightarrow f(u)f(v) \in E(H) \Leftrightarrow f(u)f(v) \notin E(\overline{H})$. So f is an isomorphism between $\overline{G}$ and $\overline{H}$. The other direction of the proof can be shown similarly.

■ **Problem 1.5** Prove or disprove. If every vertex of a simple graph G has degree 2, then G is a cycle.

Solution Not true. Two copies of K_3 is a counterexample.

■ **Problem 1.6** As in the textbook.

Solution Consider the following decomposition.



■ **Problem 1.7** Prove that a graph with more than six vertices of off degree can not be decomposed into three paths.